Gradient Boosting Classification

GB ensemble boosting Technique

In GB we take weak learners i.e.
model having high bias & low variance
This is additive modelling technique

so $\underset{M_1}{\boxed{HB}} + \underset{M_2}{\boxed{HB}} + \underset{M_3}{\boxed{HB}} \Rightarrow LB$

Example

cgpa	iq	is_placed
6.82	118	0
6.36	125	1
5.39	99	1
5.50	106	1
6.39	148	0
9.13	148	1
7.17	147	1
7.72	72	0

$f_0(x)$ $\quad$ $f_1(x)$ $\quad$ $f_2(x)$

$$F(x) = f_0(x) + f_1(x) + f_2(x)$$

Stage 1 : Simple Model

$$\log_e(\text{odds}) = \log_e\left[\frac{\#1}{\#0}\right]$$

$$= \log_e\left[\frac{5}{3}\right] = 0.51$$

∴ Initial Prediction

cgpa	iq	is_placed	pre1(log-odds)
6.82	118	0	0.510826
6.36	125	1	0.510826
5.39	99	1	0.510826
5.50	106	1	0.510826
6.39	148	0	0.510826
9.13	148	1	0.510826

7.17	147	1	0.510826
7.72	72	0	0.510826

Calculate the Probability

$$p = \frac{1}{1 + e^{-\log(\text{odds})}} = \frac{1}{1 + e^{-0.51}} = 0.62$$

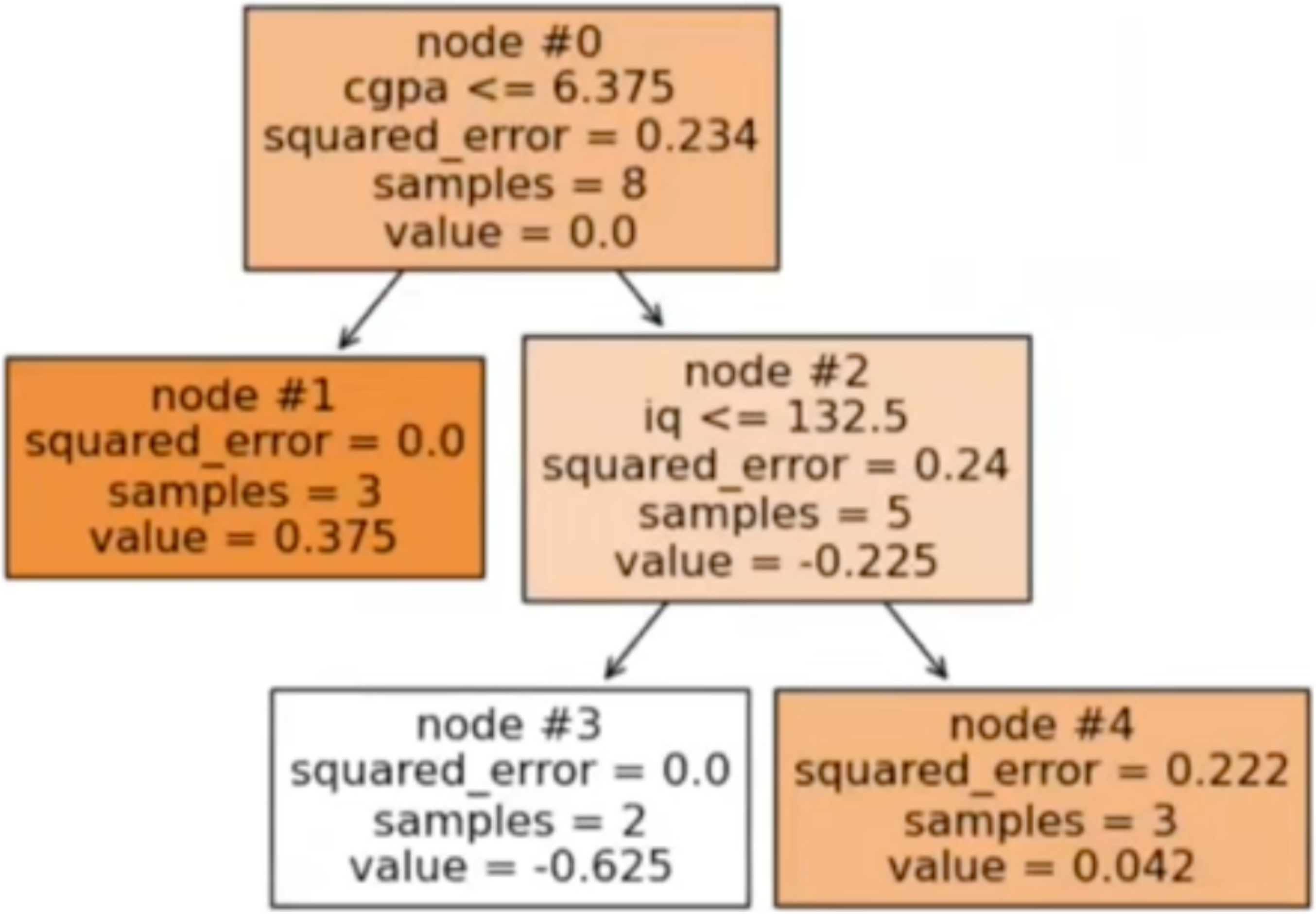
cgpa	iq	is_placed	pre1(log-odds)	pre1(probability)
6.82	118	0	0.510826	0.625
6.36	125	1	0.510826	0.625
5.39	99	1	0.510826	0.625
5.50	106	1	0.510826	0.625
6.39	148	0	0.510826	0.625
9.13	148	1	0.510826	0.625
7.17	147	1	0.510826	0.625
7.72	72	0	0.510826	0.625

Step 2 : Residual calculation

$$Res = Y - Pre1(Probability)$$

cgpa	iq	is_placed	pre1(log-odds)	pre1(probability)	res1
6.82	118	0	0.510826	0.625	-0.625
6.36	125	1	0.510826	0.625	0.375
5.39	99	1	0.510826	0.625	0.375
5.50	106	1	0.510826	0.625	0.375
6.39	148	0	0.510826	0.625	-0.625
9.13	148	1	0.510826	0.625	0.375
7.17	147	1	0.510826	0.625	0.375
7.72	72	0	0.510826	0.625	-0.625

Stage 3 - Decision Tree 1


node #0
cgpa <= 6.375
squared_error = 0.234
samples = 8
value = 0.0
node #1
squared_error = 0.0
samples = 3
value = 0.375
node #2
iq <= 132.5
squared_error = 0.24
samples = 5
value = -0.225
node #3
squared_error = 0.0
samples = 2
value = -0.625
node #4
squared_error = 0.222
samples = 3
value = 0.042

Decision Tree is plotted between column

x ((gpa, iq) & residual 1

* Log (odds) for each node

$$\frac{\sum \text{Residual}}{\sum (\text{Previous Prob} \times (1 - \text{Previous Prob}))}$$

cgpa	iq	is_placed	pre1(log-odds)	pre1(probability)	res1	leaf_entry1
6.82	118	0	0.510826	0.625	-0.625	3
6.36	125	1	0.510826	0.625	0.375	1
5.39	99	1	0.510826	0.625	0.375	1
5.50	106	1	0.510826	0.625	0.375	1
6.39	148	0	0.510826	0.625	-0.625	4
9.13	148	1	0.510826	0.625	0.375	4
7.17	147	1	0.510826	0.625	0.375	4
7.72	72	0	0.510826	0.625	-0.625	3

Above table shows which row is Falling under which leaf node.

Using above Formula calculating log(odd) For node #3

$$= \frac{-0.625 - 0.625}{0.625(1-0.625) + 0.625(1-0.625)}$$

$$= \frac{-2 \times 0.625}{0.625 \times 0.375 + 0.625 \times 0.375}$$

$$= \frac{2 \times \cancel{0.625}}{2 \times \cancel{0.625} \times 0.375} = -\frac{1}{0.375}$$

log(odds) = −2.66

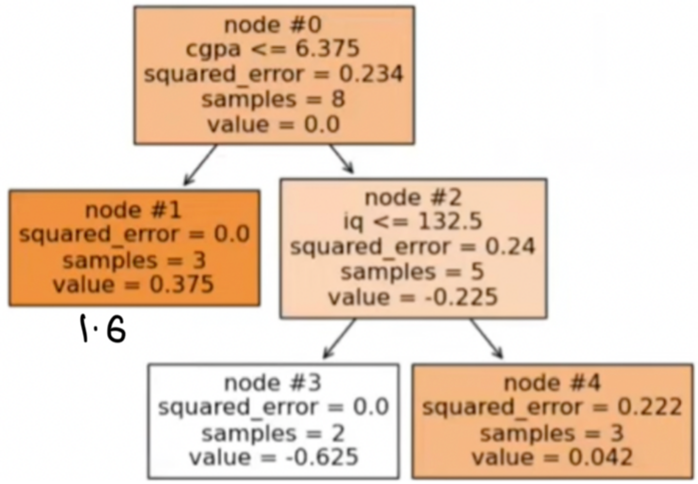
Similarly, for node #1 log(odd) = 1.6

For node #4, log(odd) = 0.18

Output → log(odds) for model 2

$f_0(x) +$

$\Rightarrow 0.51 +$



1.6   −2.66   0.18

for row 1, Pre2(log-odds) $= 0.51 + (-2.66)$

$= -2.15$

Residual calculation for M_2.

which is nothing but probability

$$p = \frac{1}{1 + e^{-\log(\text{odds})}}$$

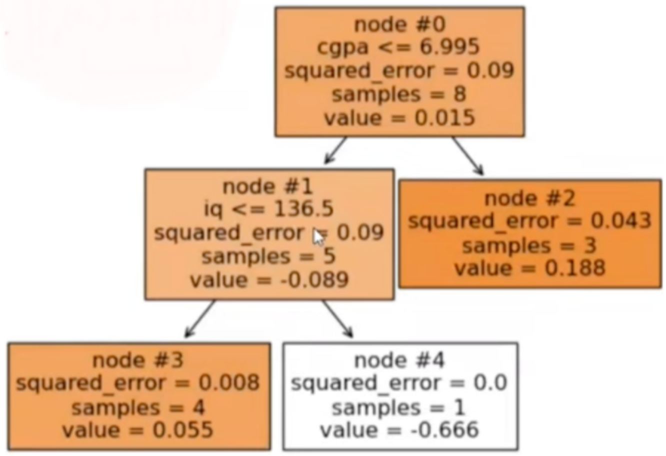
cgpa	iq	is_placed	pre1(log-odds)	pre1(probability)	res1	leaf_entry1	pre2(log-odds)	pre2(probability)
6.82	118	0	0.510826	0.625	-0.625	3	-2.159174	0.103477
6.36	125	1	0.510826	0.625	0.375	1	2.110826	0.891951
5.39	99	1	0.510826	0.625	0.375	1	2.110826	0.891951
5.50	106	1	0.510826	0.625	0.375	1	2.110826	0.891951
6.39	148	0	0.510826	0.625	-0.625	4	0.690826	0.666151
9.13	148	1	0.510826	0.625	0.375	4	0.690826	0.666151
7.17	147	1	0.510826	0.625	0.375	4	0.690826	0.666151
7.72	72	0	0.510826	0.625	-0.625	3	-2.159174	0.103477

Residual calculation

$$y - [f_0(x_i) + f_1(x_i)]$$

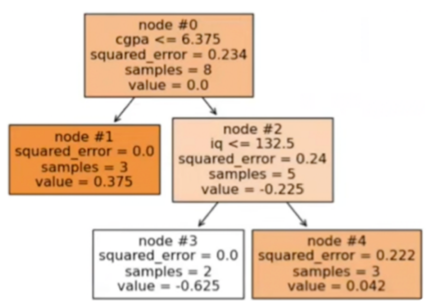
cgpa	iq	is_placed	pre1(log-odds)	pre1(probability)	res1	leaf_entry1	pre2(log-odds)	pre2(probability)	res2
6.82	118	0	0.510826	0.625	-0.625	3	-2.159174	0.103477	-0.103477
6.36	125	1	0.510826	0.625	0.375	1	2.110826	0.891951	0.108049
5.39	99	1	0.510826	0.625	0.375	1	2.110826	0.891951	0.108049
5.50	106	1	0.510826	0.625	0.375	1	2.110826	0.891951	0.108049
6.39	148	0	0.510826	0.625	-0.625	4	0.690826	0.666151	-0.666151
9.13	148	1	0.510826	0.625	0.375	4	0.690826	0.666151	0.333849
7.17	147	1	0.510826	0.625	0.375	4	0.690826	0.666151	0.333849
7.72	72	0	0.510826	0.625	-0.625	3	-2.159174	0.103477	-0.103477

stage 4: Decision Tree 2



Repeating same steps for Decision Tree 2

$f_0(x) +$



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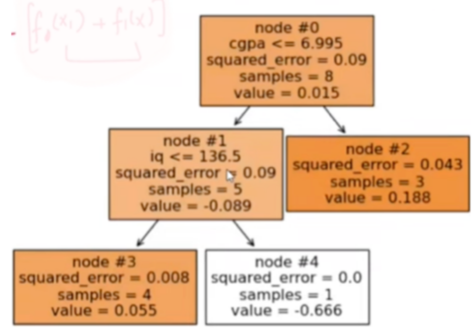

$$\log(\text{odds}) = \left[\frac{\sum \text{Residual}}{\sum(\text{Prev. Prob} \times (1-\text{Prev Prob}))}\right]$$

$$\log(\text{odds})\ \#2 = 1.56$$

$\#3 = 1.09$

$\#4 = -2.94$

Calculating Probability

$$p = \frac{1}{1 + e^{-\log(odds)}}$$

cgpa	iq	is_placed	pre1(log-odds)	pre1(probability)	res1	leaf_entry1	pre2(log-odds)	pre2(probability)	res2	leaf_entry2	pre3(log-odds)	pre3(probability)
6.82	118	0	0.510826	0.625	-0.625	3	-2.159174	0.103477	-0.103477	3	-1.068349	0.255717
6.36	125	1	0.510826	0.625	0.375	1	2.110826	0.891951	0.108049	3	3.201651	0.960896
5.39	99	1	0.510826	0.625	0.375	1	2.110826	0.891951	0.108049	3	3.201651	0.960896
5.50	106	1	0.510826	0.625	0.375	1	2.110826	0.891951	0.108049	3	3.201651	0.960896
6.39	148	0	0.510826	0.625	-0.625	4	0.690826	0.666151	-0.666151	4	-1.798349	0.142052
9.13	148	1	0.510826	0.625	0.375	4	0.690826	0.666151	0.333849	2	2.251651	0.904793
7.17	147	1	0.510826	0.625	0.375	4	0.690826	0.666151	0.333849	2	2.251651	0.904793
7.72	72	0	0.510826	0.625	-0.625	3	-2.159174	0.103477	-0.103477	2	-0.598349	0.354722

Prediction

$$\{72, 100\} \Rightarrow \{0, 1\}$$

$$0.51 + (-2.66) + 1.56 = -0.59$$

$$\log(\text{odds}) = -0.59$$

$$\text{Prob} = \frac{1}{1 + e^{-0.59}} = 0.35$$

if Threshold $= 0.5 \Rightarrow$ output will be 0